

## **PART A: Short Answer Questions**

### **1. What is system context?**

System context is the environment surrounding a system, including all external entities that interact with it.

### **2. List TWO aspects that influence system context.**

- Stakeholders (users, organizations)
- External systems and regulations

### **3. State the purpose of defining system and context boundaries.**

To identify what is inside the system and what is outside the system.

### **4. Define system boundary.**

The system boundary defines the functions and components that belong to the system.

### **5. Define context boundary.**

The context boundary separates the system and its environment from unrelated external entities.

### **6. What is the consequence of incorrect system context?**

It may lead to incorrect requirements and system failures.

### **7. What are sources and sinks?**

Sources provide data to the system, while sinks receive data from the system.

### **8. What is the grey zone between system and system context?**

The grey zone refers to components whose ownership or responsibility is unclear between the system and its environment.

### **9. Name TWO techniques to document system context.**

- Context Diagrams
- Use Case Diagrams

### **10. What is tool support in Requirements Engineering?**

Tool support refers to software tools used to assist requirements engineering activities.

### **11. Give TWO examples of general tool support.**

- Microsoft Word
- Microsoft Excel

### **12. State ONE advantage of using general development tools for requirements management.**

They are easy to use and widely available.

### **13. What are modelling tools used for?**

Modelling tools are used to create visual representations of systems and requirements.

### **14. State TWO properties of modelling tools.**

- Graphical representation of requirements

- Support for analysis and communication

**15. List TWO categories of requirements management tools.**

- General-purpose tools
- Specialized requirements management tools

**PART B: Case Study Questions**

**Case Study: Student Management System**

**1. Identify THREE elements of the system context.**

- Students
- Lecturers
- Payment System

**2. Differentiate between system and system context.**

The **system** is the Student Management System itself that manages student information and services. The **system context** consists of external entities such as students, lecturers, payment systems, and university regulations that interact with the system.

**3. Give ONE example of system boundary.**

Student registration and course enrollment functions are inside the system boundary.

**4. Give ONE example of context boundary.**

The external payment processing service is outside the system boundary but within the system context.

**5. Explain the grey zone in this case.**

Payment processing may be partially controlled by the university and partially by an external provider, making responsibility unclear.

**6. Identify ONE source and ONE sink.**

- Source: Student submitting registration information.
- Sink: Lecturer receiving student course information.

**7. Suggest ONE suitable general tool and justify.**

**Microsoft Word** can be used because it allows requirements to be documented, edited, and shared easily.

**8. Explain how modelling tools can be used in this system.**

Modelling tools can create use case diagrams, context diagrams, and process models to visualize system requirements and interactions.

**9. Why is traceability important?**

Traceability helps track requirements from their origin through development and testing, ensuring all requirements are implemented correctly.

**10. Evaluate a tool using ONE evaluation view.**

**Usability View:** Microsoft Word is easy to learn and use, making it suitable for documenting and managing requirements efficiently.